

## Tude Dude Assignment 2

**GitHub Repo Link:** [Harshit765G4/TudeDude-Assignments](https://github.com/Harshit765G4/TudeDude-Assignments)

### 1. Grade Checker

Take a score as input and print the grade based on the following:

**90+ : "A"**

**80-89 : "B"**

**70-79 : "C"**

**60-69 : "D"**

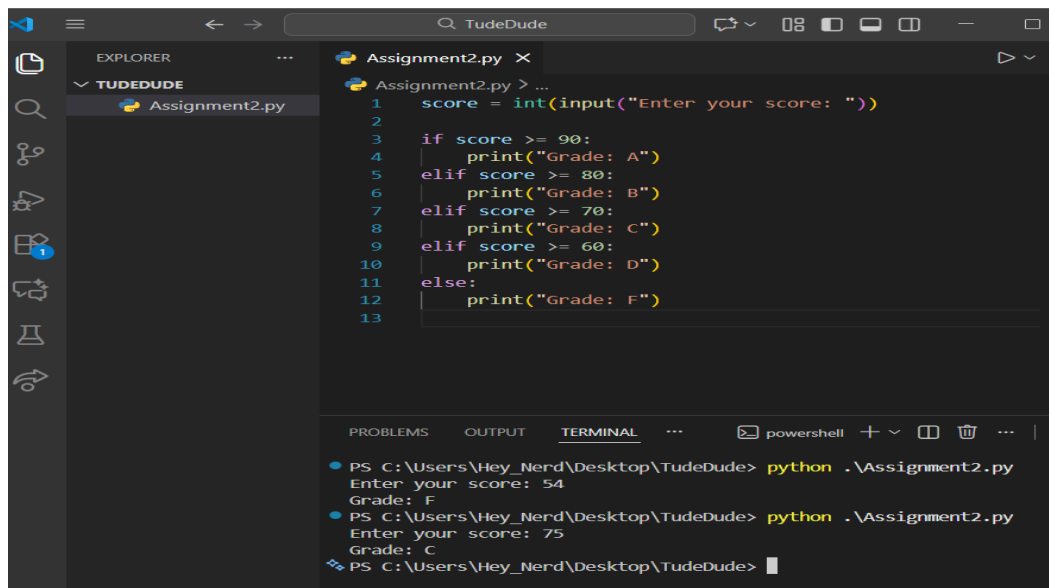
**Below 60 : "F"**

here we used a basic if else statement to carry out marks and all.

**Solution:**

```
score = int(input("Enter your score: "))  
if score >= 90:  
    print("Grade: A")  
elif score >= 80:  
    print("Grade: B")  
elif score >= 70:  
    print("Grade: C")  
elif score >= 60:  
    print("Grade: D")  
else:  
    print("Grade: F")
```

This takes marks as input and uses if-elif-else conditions to check the score range.



The screenshot shows a Visual Studio Code editor window with a file named 'Assignment2.py'. The code in the file is as follows:

```
1 score = int(input("Enter your score: "))
2
3 if score >= 90:
4     print("Grade: A")
5 elif score >= 80:
6     print("Grade: B")
7 elif score >= 70:
8     print("Grade: C")
9 elif score >= 60:
10    print("Grade: D")
11 else:
12    print("Grade: F")
13
```

Below the code editor, the 'TERMINAL' panel shows the execution of the script in a PowerShell environment. The first run shows an input of 54 resulting in a grade of F. The second run shows an input of 75 resulting in a grade of C.

```
PS C:\Users\Hey_Nerd\Desktop\TudeDude> python .\Assignment2.py
Enter your score: 54
Grade: F
PS C:\Users\Hey_Nerd\Desktop\TudeDude> python .\Assignment2.py
Enter your score: 75
Grade: C
PS C:\Users\Hey_Nerd\Desktop\TudeDude>
```

## 2 Student Grades

Create a dictionary where the keys are student names and the values are their grades. Allow the user to:

Add a new student and grade.

Update an existing student's grade.

Print all student grades.

**Solution:**

```
students = {}

while True:

    print("\n1. Add Student")

    print("2. Update Student Grade")

    print("3. Display All Grades")

    print("4. Exit")

    choice = input("Enter choice: ")
```

```
if choice == "1":
    name = input("Enter student name: ")
    grade = input("Enter grade: ")
    students[name] = grade
    print("Student added successfully.")

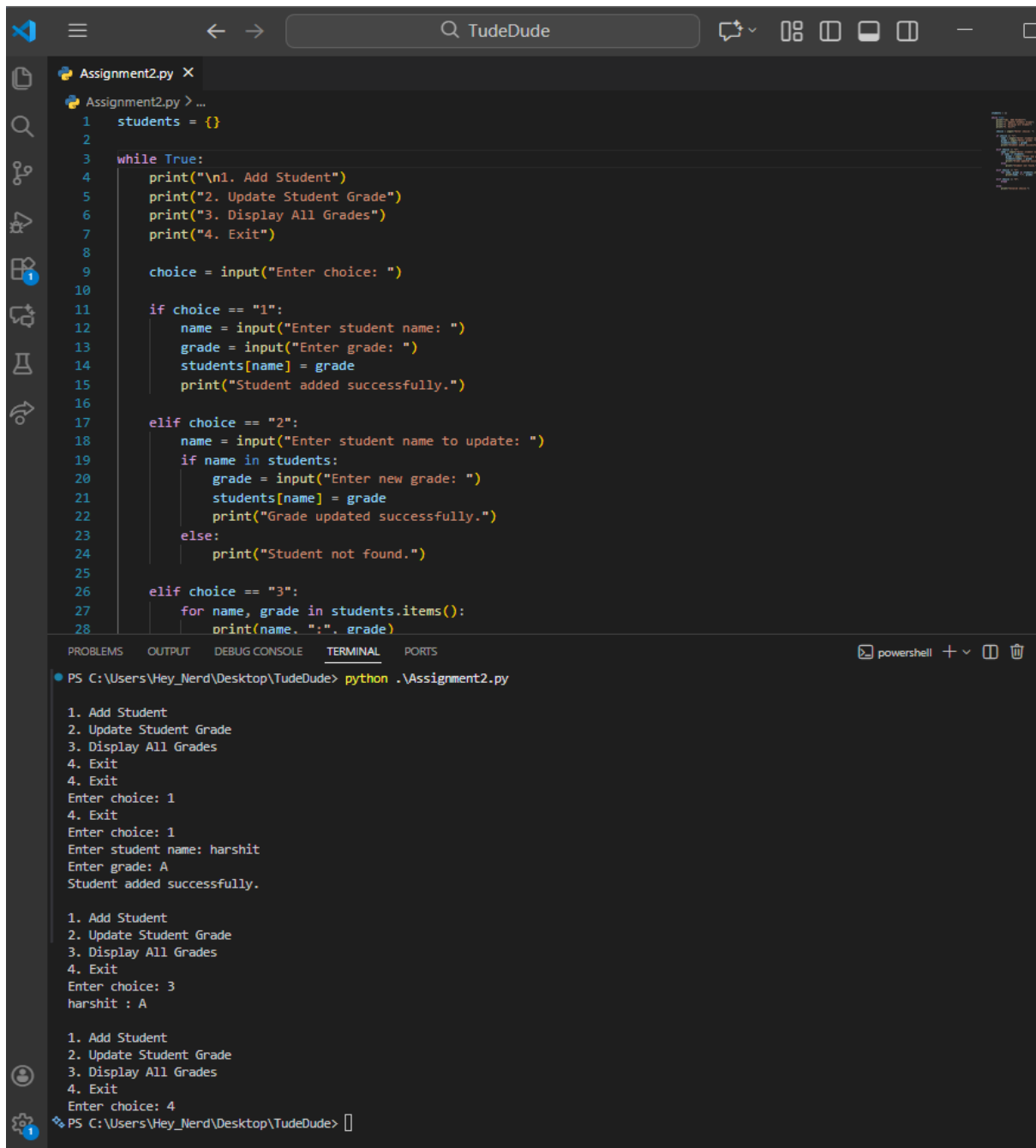
elif choice == "2":
    name = input("Enter student name to update: ")
    if name in students:
        grade = input("Enter new grade: ")
        students[name] = grade
        print("Grade updated successfully.")
    else:
        print("Student not found.")

elif choice == "3":
    for name, grade in students.items():
        print(name, ":", grade)

elif choice == "4":
    break

else:
    print("Invalid choice.")
```

The allows adding, updating, and displaying grades using if-else logic and dictionary operations.



The image shows a Visual Studio Code editor window with a file named `Assignment2.py` open. The code is a Python script that implements a menu-driven program for managing student grades. The script uses a dictionary to store student names and their grades, and a `while` loop to repeatedly show a menu of options to the user. The options are: 1. Add Student, 2. Update Student Grade, 3. Display All Grades, and 4. Exit. The user is prompted to enter a choice, and the program executes the corresponding function. For example, when the user enters 1, the program prompts for a student name and a grade, and then adds the student to the dictionary. When the user enters 3, the program displays all the students and their grades.

```
1 students = {}
2
3 while True:
4     print("\n1. Add Student")
5     print("2. Update Student Grade")
6     print("3. Display All Grades")
7     print("4. Exit")
8
9     choice = input("Enter choice: ")
10
11     if choice == "1":
12         name = input("Enter student name: ")
13         grade = input("Enter grade: ")
14         students[name] = grade
15         print("Student added successfully.")
16
17     elif choice == "2":
18         name = input("Enter student name to update: ")
19         if name in students:
20             grade = input("Enter new grade: ")
21             students[name] = grade
22             print("Grade updated successfully.")
23         else:
24             print("Student not found.")
25
26     elif choice == "3":
27         for name, grade in students.items():
28             print(name, ":", grade)
```

The terminal output shows the program being executed in a PowerShell window. The user enters the command `python .\Assignment2.py`. The program then displays the menu and the user interacts with it. The first interaction shows the user entering choice 1, then a student name 'harshit' and a grade 'A', and the program prints 'Student added successfully.'. The second interaction shows the user entering choice 3, and the program displays the list of students and their grades: 'harshit : A'. The third interaction shows the user entering choice 4, and the program ends.

### 3. Write to a File

Write a program to create a text file and write some content to it.

Using file functions like write and open.

**Solution:**

```
file = open("sample.txt", "w")

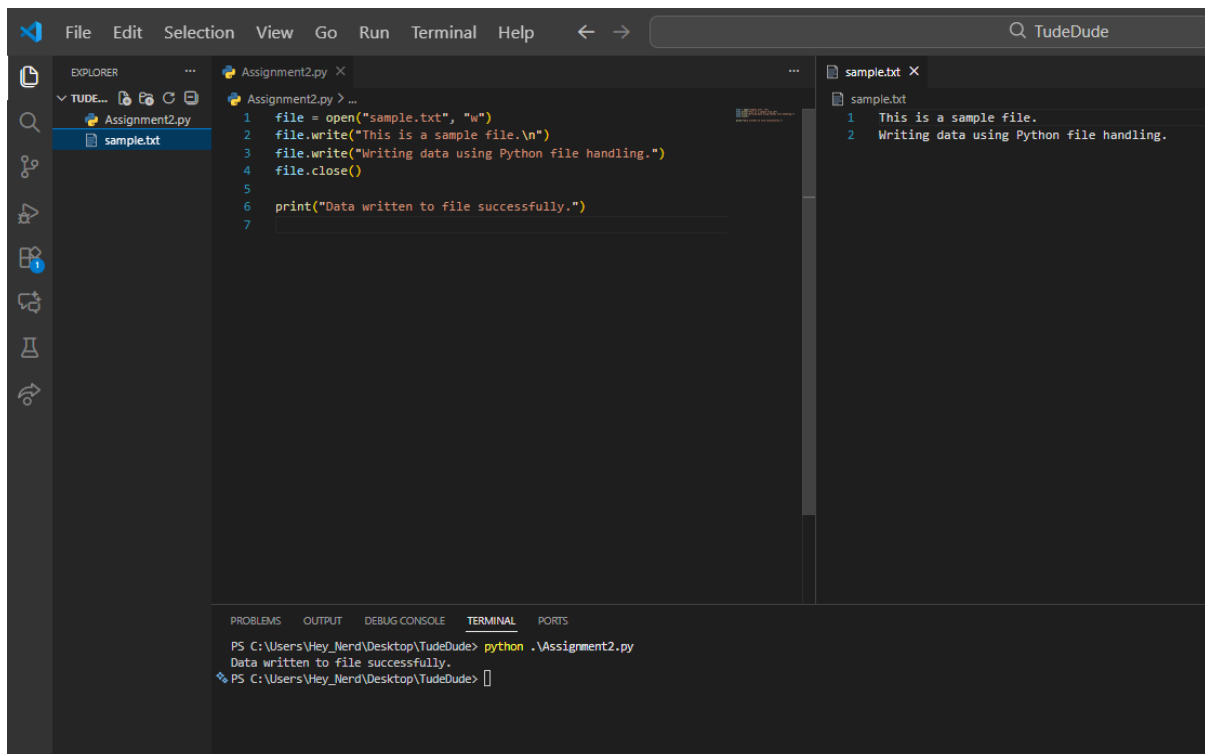
file.write("This is a sample file.\n")

file.write("Writing data using Python file handling.")

file.close()


print("Data written to file successfully.")
```

The file is opened in write (w) mode.



#### 4. Read from a File

We used open in read mode and file.read to read and print to display.

**Solution:**

The file is opened in read (r) mode.

```
file = open("sample.txt", "r")
```

```
content = file.read()
```

```
print(content)
```

```
file.close()
```

